Chapter 2 — Bacteria and Viruses

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Main component of bacterial cell wall is: (a) Cellulose (b) Peptidoglycan (c) Chitin (d) Silica

2. Gram-negative outer membrane contains: (a) Lipopolysaccharides (b) Peptidoglycan (c) Cellulose (d) Chitin

3. Bacterial plasma membrane infoldings are: (a) Mesosomes (b) Plasmids (c) Capsules (d) Ribosomes

4. Extra-chromosomal circular DNA in bacteria is: (a) Plasmids (b) Mesosomes (c) Nucleoids (d) Capsules

5. Endospores are mainly for: (a) Reproduction (b) Survival in harsh conditions (c) Photosynthesis (d) Conjugation

6. Bacteria with one polar flagellum are: (a) Peritrichous (b) Monotrichous (c) Lophotrichous (d) Amphitrichous

7. Intestinal normal flora produce: (a) Antibodies (b) Insulin (c) Vitamin C (d) Vitamin K and B12

8. Protein coat of a virus is: (a) Nucleoid (b) Envelope (c) Capsid (d) Capsomere

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Differentiate between the cell walls of Gram-positive and Gram-negative bacteria.

2. How do endospores help bacteria survive unfavourable conditions?

3. Differentiate between flagella and pili in bacteria.

4. How does normal flora prevent colonization by pathogens?

5. Why are viruses considered acellular?

6. What is the role of tail fibres in a bacteriophage?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Describe the structure of a generalized bacterial cell. **(4)**

(b) Explain the process of endospore formation in bacteria. **(4)**

Q.4 (a) Describe the arrangement and structure of bacterial flagella. **(4)**

(b) Describe the structure of HIV and explain why it is called a retrovirus. **(4)**

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